Principles of Economics

Final termTopic 3

Business Cycle and Unemployment

Macroeconomic Instability

- Economic Growth
- Business Cycle
- Unemployment
- Inflation

Economic Growth

- Economists define and measure economic growth as either:

→ An increase in real GDP occurring over some time period

→ An increase in real GDP per capita occurring over some time period.

Economic Growth

- If real GDP was $ 200 billion in some country last year and $ 210 billion this year, the rate of growth would be 5 percent.

- Rule of 70:

Approximate number of years required to double real GDP = (70 / annual percentage rate of growth)

Example : Let’s say : RGDP 2021= 80 million dollars

- RGDP 2022= 100 million dollars
- Question : How many years approx. it will take to double the RGDP?
- Solution: rate of growth = 100-80/80*100= 25%
- Approx years = 70/25 = 2.8 years

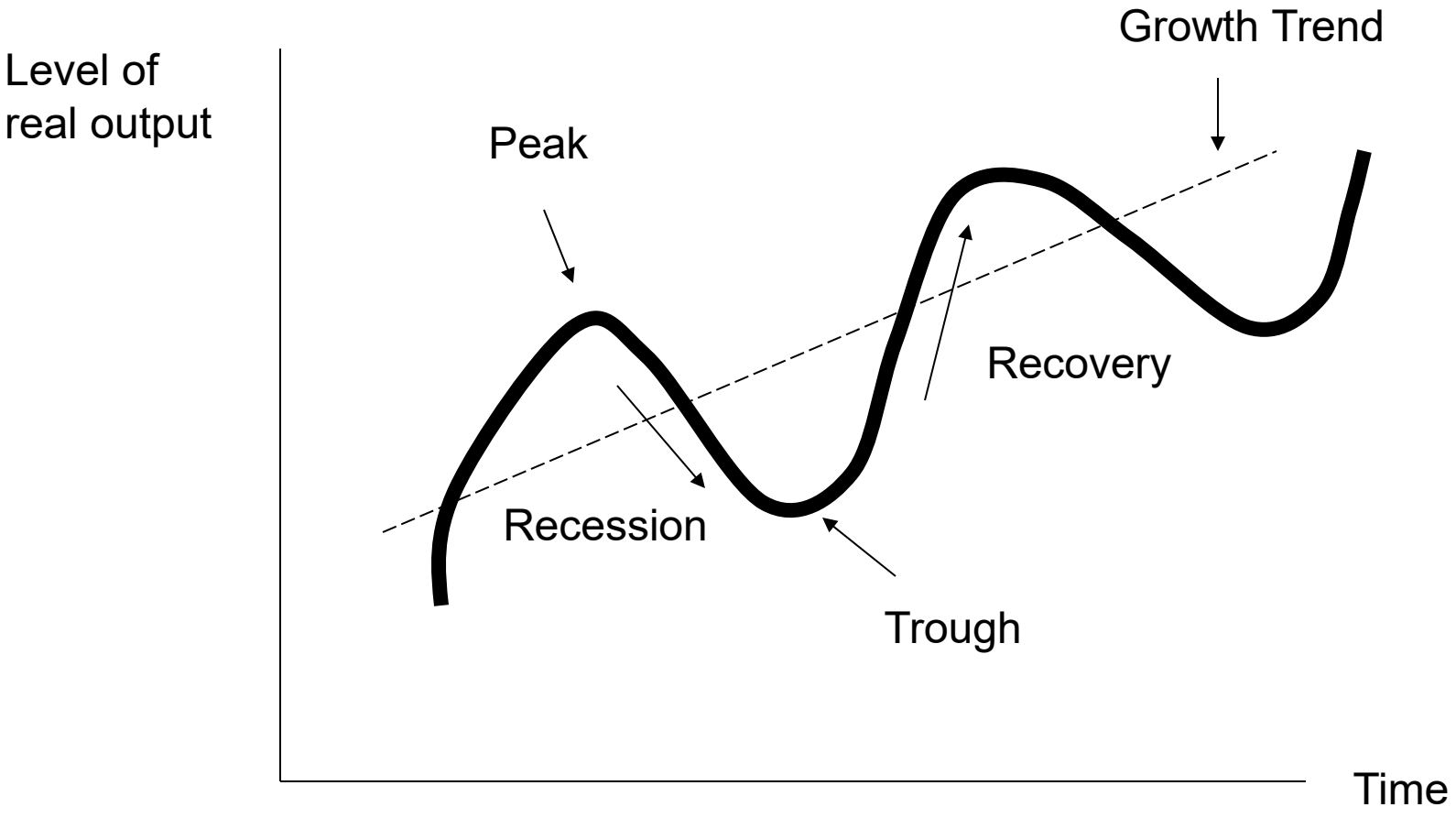
The Business Cycle

- The term business cycle refers to alternating rises and declines in the level of economic activity.
- Decline in real GDP leads to significant increases in unemployment.
- Rapid economic growth has been marred by rapid inflation.

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The Business Cycle



Phases of the Business Cycle

- Peak:

  → at a peak business activity has reached a temporary maximum.

  → the economy is at full employment.

  → very close to full productivity.

- The price level is likely to rise during this phase.

Phases of the Business Cycle

- Recession:

  → a period of decline in total output, income, employment, and trade.

  → wide spread contraction of business activity.

- The price level is likely to fall.

Phases of the Business Cycle

- Trough:

  → output and employment bottom out at their lowest levels.

  → the trough phase may be either short lived or quite long.

Phases of the Business Cycle

- Recovery:

  → in the recovery phase, output and employment rise toward full employment.

- The price level may begin to rise before full employment and full capacity production return.

Unemployment

- The labor force consists of people who are able and willing to work.
- Both those who are employed and those who are unemployed but actively seeking work are counted as being in the labor force.

BLS divides population into 3 groups:

- **Employed**: paid employees, self-employed, and unpaid workers in a family business
- **Unemployed**: people who were available and not working but who have looked for work during previous 4 weeks
- **Not in the labor force**: those who do not fit either any of the above two categories. i.e. full-time student, homemaker, or retiree

The **labor force** is the total # of workers, including the employed and unemployed.

Labor Force Statistics

- **Unemployment rate** ("u-rate"): % of the labor force that is unemployed

- Unemployment rate=$\frac{number\ of\ unemployed}{labor\ force}$×100
- **Labor force participation rate**:
  % of the adult population that is in the labor force
- Labor Force Participation Rate =$\frac{Labor\ Force}{Adult\ Population}$×100

Compute the labor force, u-rate, adult population, and labor force participation rate using this data:

Adult population of the U.S. by group, June 2008	
# of employed	145.9 million
# of unemployed	8.5 million
not in labor force	79.2 million

Answers

Labor force = employed + unemployed

= 145.9 + 8.5

= 154.4 million

U-rate = 100 x (unemployed)/(labor force)

= 100 x 8.5/154.4

= 5.5%

Answers

Population = labor force + not in labor force

= 154.4 + 79.2

= 233.6

LF partic. rate = 100 x (labor force)/(population)

= 100 x 154.4/233.6

= 66.1%

Unemployment

- The unemployment rate is the percentage of the labor force unemployed:

Unemployment rate= (unemployed / labor force) X 100

Types of Unemployment

- There are three types of unemployment:

  → Frictional Unemployment
  → Structural Unemployment
  → Cyclical Unemployment

Frictional Unemployment

- Frictional unemployment consisting of search unemployment and wait unemployment.
- The word frictional implies that the labor market does not operate perfectly in matching workers and jobs.

Why Frictional Unemployment is inevitable ?

Sectoral shifts

- These shifts are changes in the composition of demand across industries or regions of the country.
- Such shifts displace some workers, who must search for new jobs appropriate for their skills & tastes.
- The economy is always changing, so some frictional unemployment is inevitable.

Cyclical Unemployment

- Cyclical unemployment is caused by a decline in total spending and is likely to occur in the recession phase of the business cycle.

Structural Unemployment

- Structural unemployment results because the composition of the labor force does not respond immediately or completely to the new structure of job opportunities.
- Structurally unemployed workers find it hard to obtain new jobs without retraining, gaining additional education, or relocating.

Why there is Structural unemployment?

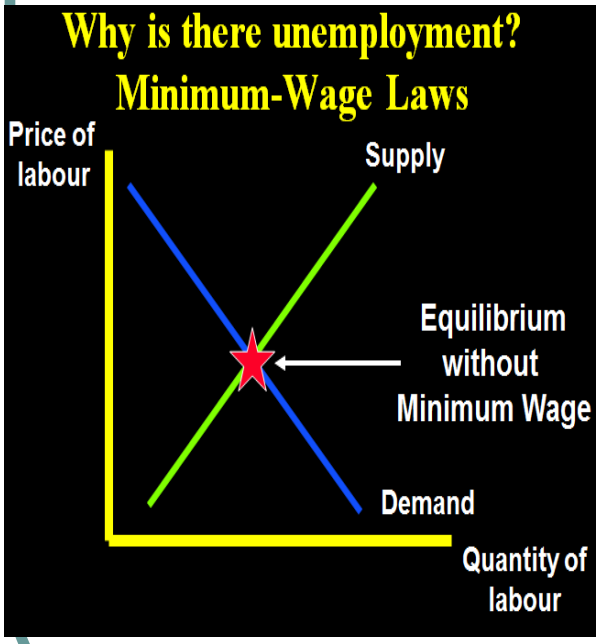
Structural Unemployment occurs when wage is kept above equilibrium

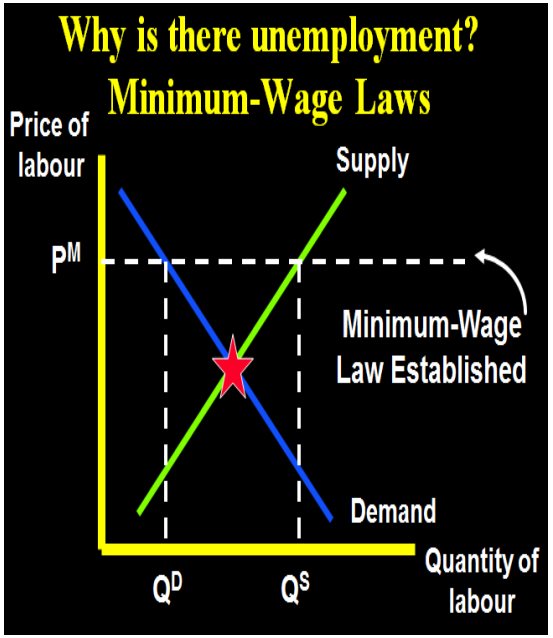
There are three reasons for this→

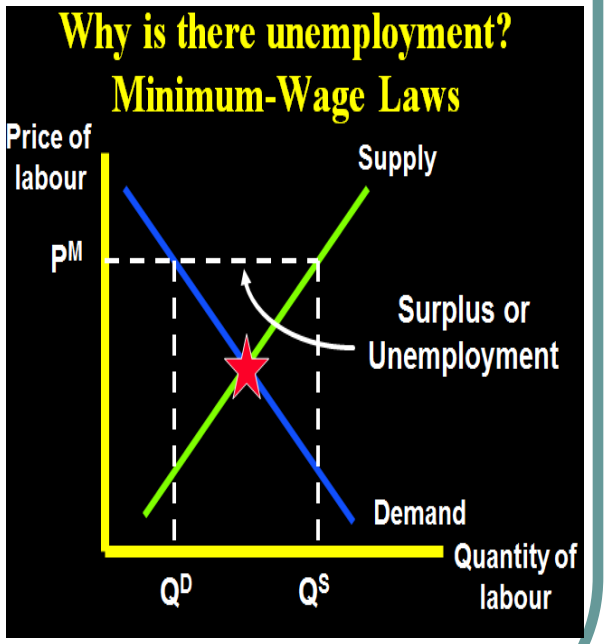
1. Minimum-Wage Laws
2. Unions
3. Efficiency Wages

Minimum Wage Laws

- When a minimum-wage law forces the wage to remain above the level that balances supply and demand, it creates a *surplus of labor*.







Union

- A *union* is a worker association that bargains with employers over wages and working conditions.

- A union is a type of cartel.
- Unions exert their market power to negotiate higher wages for workers.

- The typical union worker earns 20% higher wages and gets more benefits than a nonunion worker for the same type of work.



Employment situation under union

- When unions raise the wage above equilibrium, quantity of labor demanded falls than quantity supply of labor and hence unemployment results.
- “Insiders” → workers who remain employed, they are better off
- “Outsiders”→ workers who lose their jobs, they are worse off
- Some outsiders go to non-unionized labor markets, which increases labor supply and reduces wages in those markets.

Are unions good or bad for the economy?

Are unions good or bad?

- ***Critics:***
  Unions are cartels. They raise wages above equilibrium, which causes unemployment and/or depresses wages in non-union labor markets.

- ***Advocates:***
  Unions counter the market power of large firms, make firms more responsive to workers' concerns

Efficiency Wages

- The theory of **efficiency wages**:
- Firms operate more efficiently if wages are *above* the equilibrium level.
- Firms voluntarily pay above-equilibrium wages to boost worker productivity.
- But in the case of unions and minimum wage laws, firms that pay above-equilibrium wages do so involuntarily.



Four reasons why firms might pay efficiency wages:

1. **Worker health**
   Paying higher wages allows workers to eat better, makes them healthier, more productive.
2. **Worker turnover**
   Hiring & training new workers is costly.
   Paying high wages to existing workers gives more incentive to stay, reduces turnover.
3. **Worker quality**
   Offering higher wages attracts better job applicants, increases quality of the firm's workforce.
4. **Worker effort**
   If market wage is above equilibrium wage, there aren't enough jobs to go around, so workers have more incentive to work not shirk.